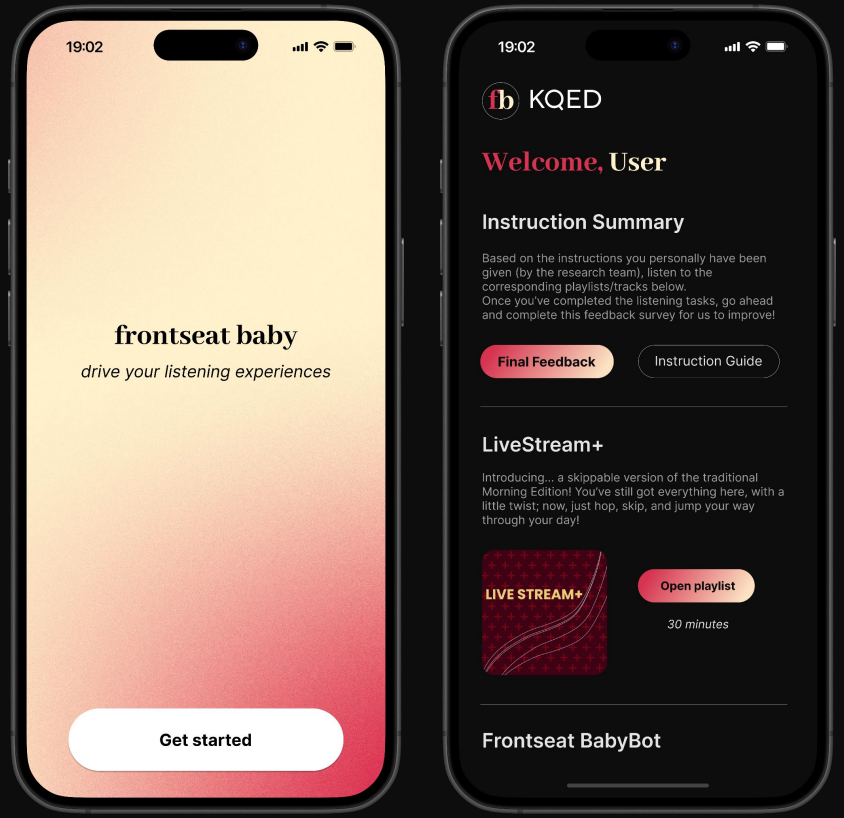
EXPLORING COMPUTATIONAL JOURNALISM
WEEK 10, PHASE 3.0 DEMO
tuesday, 03.10.26

Frontseat Baby

Drive your listening experiences

JOSH B., ELLA J., ANANYA N., JAMES V.



Meet the Team

Josh Bowden

Computer Science

Ella Jackson

Data Journalism

Ananya Navale

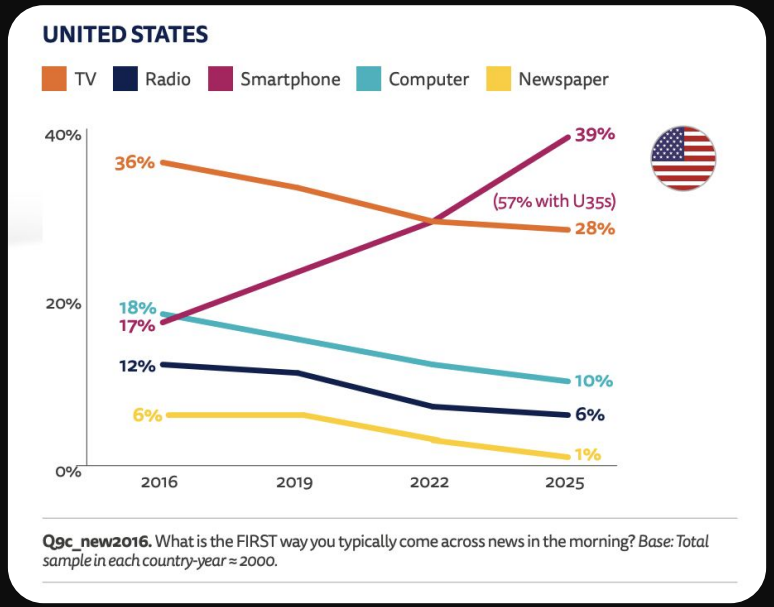
Computer Science

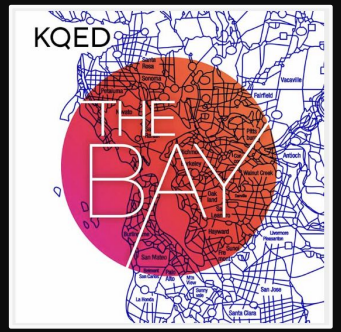
James Varah

Computer Science

Problem & Solution Overview

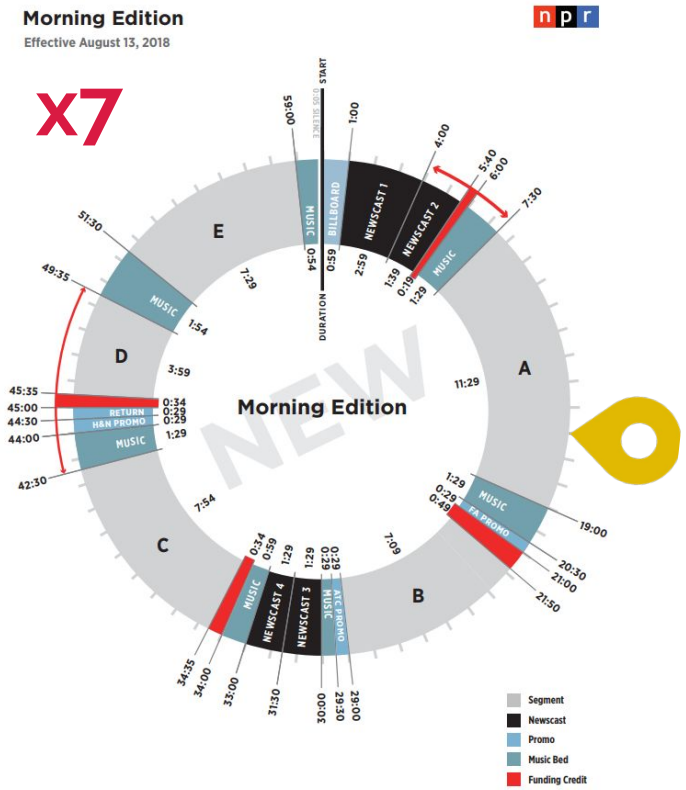
News Habits are Changing.





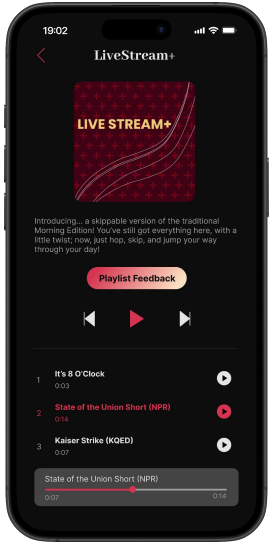
Key content is being left on the table.

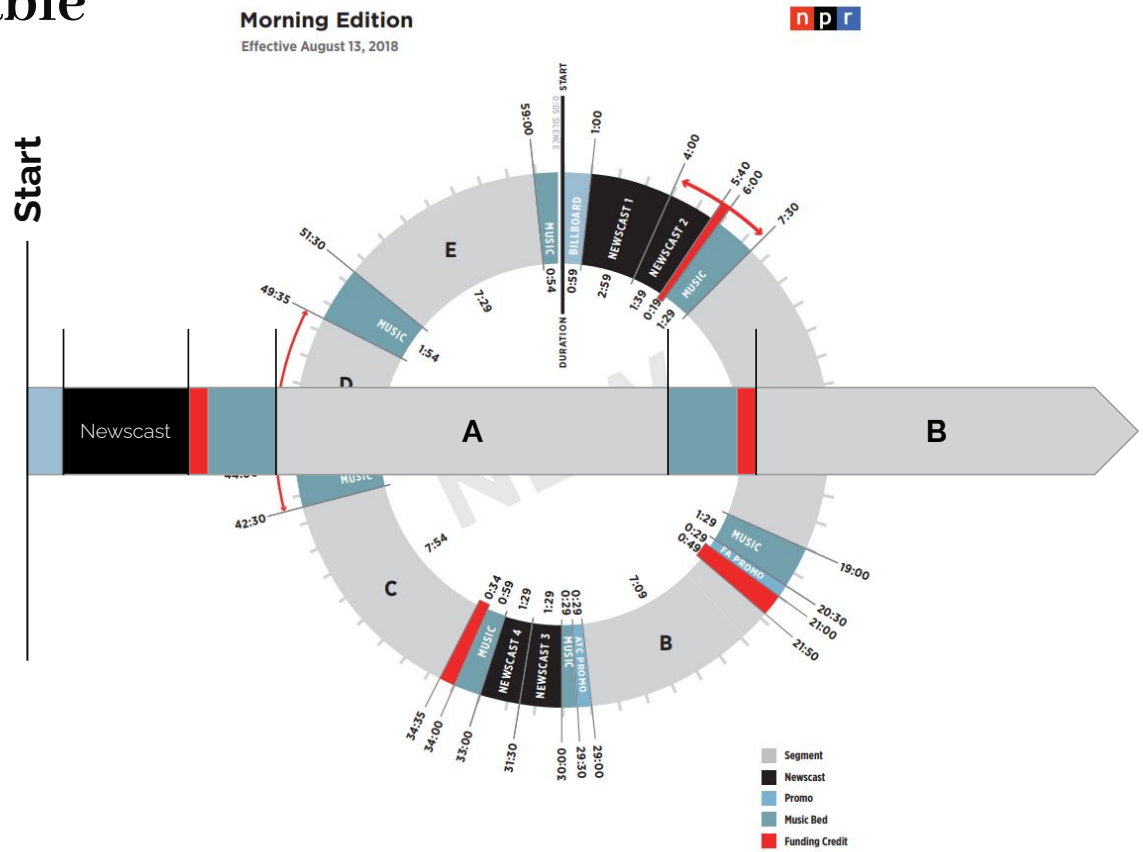
Today's Radio is Built on Old Infrastructure



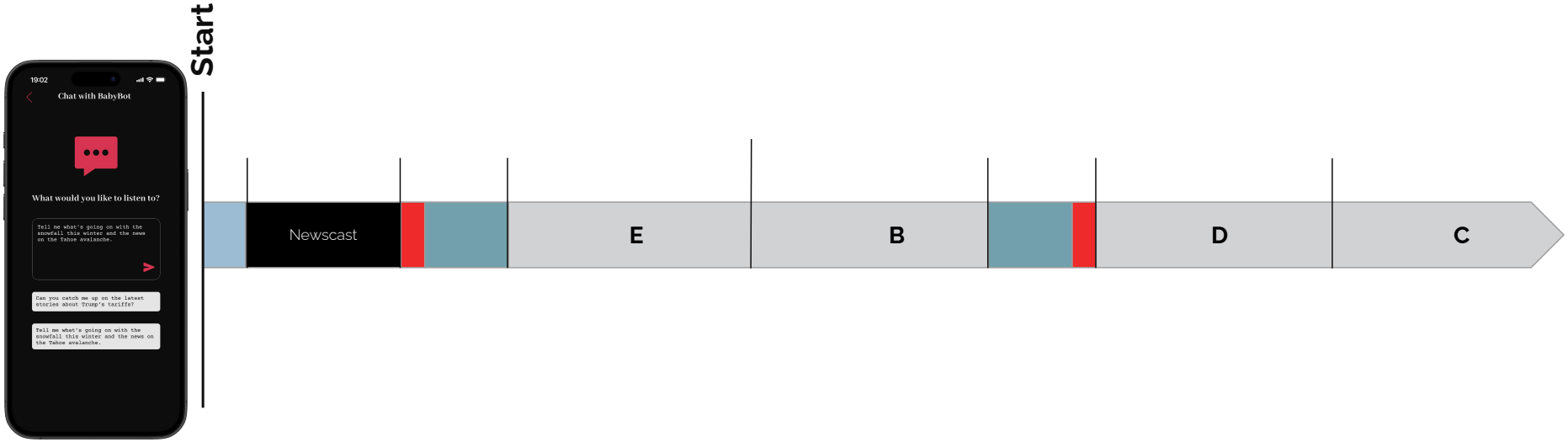
1. Many KQED clips are not **saved** and don't show up online.
2. No way to **search** and **share**.
3. There's no way to control when you **join** the live stream or **skip** old content.
4. There's no way to **personalize** live stream content.

Solution 1 → Skippable





Solution 2 → Personalized



User Study & Results

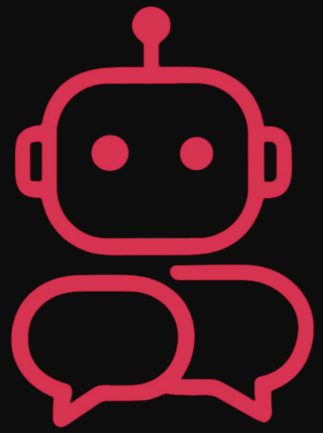
A Two-Part Approach

LiveStream+

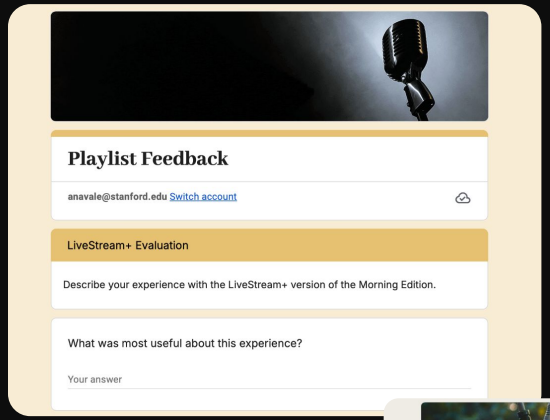
A **skippable playlist** of traditional Morning Edition **clips** that allows the user to bypass stories, ads, traffic/weather reports, etc.

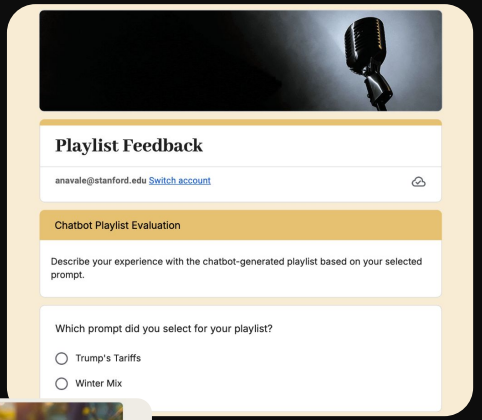
Chatbot-"generated" Playlist

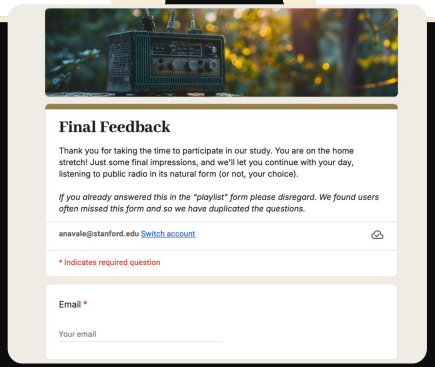
A compilation of **manually clipped** tracks from one episode of Morning Edition and based on a **personalized prompt**, assembled into Morning Edition-like playlist

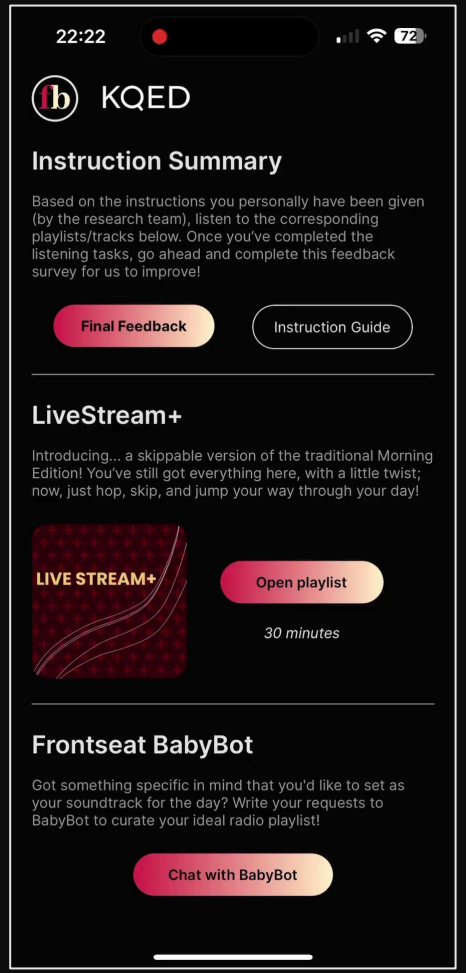


UI Prototype Demo

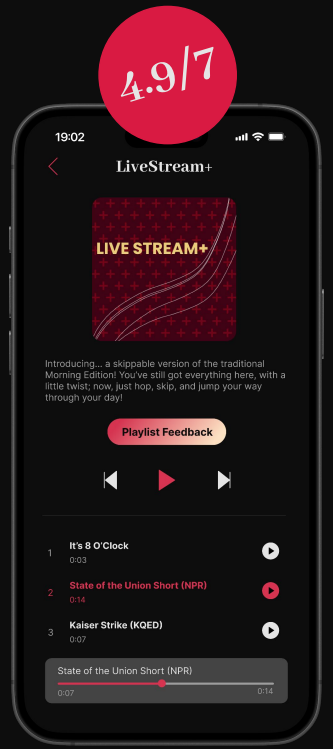








Results

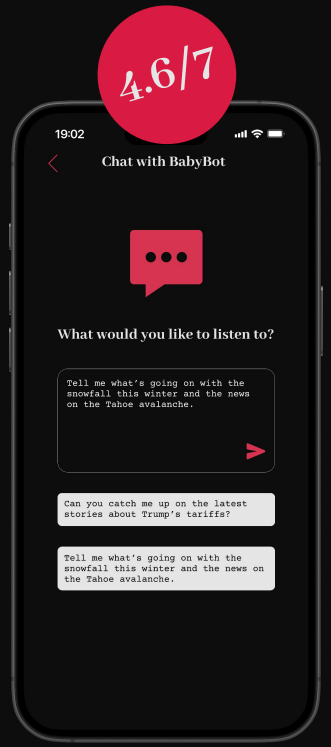


Likes

- **Skipping!**
- Good **balance** between autopilot and control

Dislikes

- No need for ads...
- Excessive segmenting



Likes

- More tailored to **personal interests**!
- **Skipped less** content
- Liked that the "Winter Mix" ended up being **more local**

Dislikes

- Content disconnected from given prompt

Key Findings

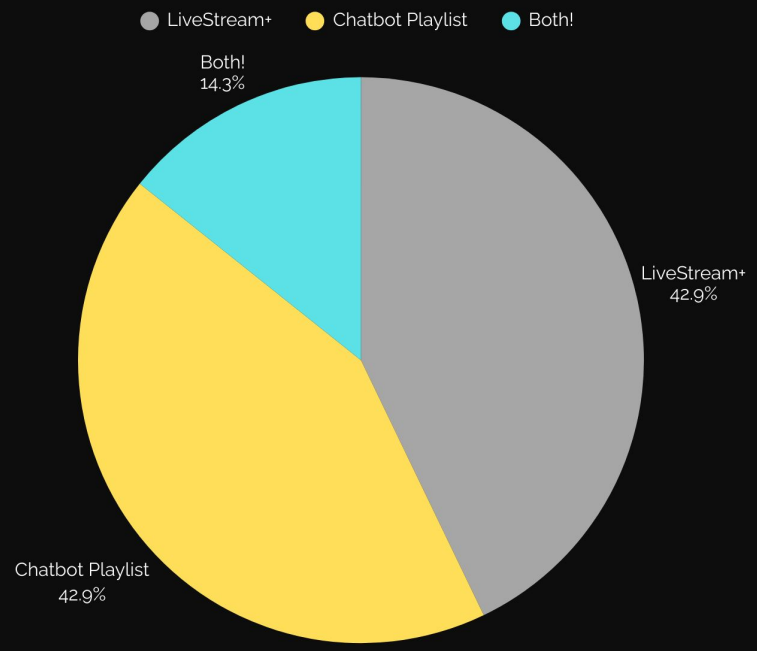
The Personalization Dilemma

- Randomness of discovery of regular radio is still exciting!
- But curating a playlist has its benefits… it is a **personal preference**

Key Risks

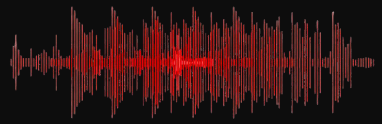
- Some users wanted **no** LLM involvement, some wanted **more** LLM involvement
- AI use brings **reputational risk**
- Losing users on **irrelevance** vs. creating echo chambers with **excessive personalization**

If you could only pick one product to keep using, which would you pick?



Technical Research & Accuracy Testing

Automated Radio to Listening Experience Pipeline



+ metadata

Transcription

Speaker Detection

Challenge: Precise Transcript-Audio Syncing

Story Segmentation and Classification

Challenge: Separate Editor-defined Stories

Story Embedding Database

Challenge: Retrieve stories with natural language

Skipping Stories

User Personalization

Challenge: Great user experience

Is it possible to automate making personalized playlists?

For user test, creating **3 playlists** required **10 hours** of manual work. Why?

1) To skip stories:

Need a recorded feed, and segment start/end times

2) To make personalized playlists

Need both of above, and way of searching archive for topics of interest

Audio Segment Search and Retrieval

1. Search database of KQED radio segments with a question

```
jamesvarah@DN51u5fj ~ % python search_stories.py "Can you catch me up on
 the latest stories about Trump's tariffs?"
Searching for: 'Can you catch me up on the latest stories about Trump's
tariffs?'...
```

Search query (yellow)

Audio Segment Search and Retrieval

2. Top 3 results for short (0-5 min), medium (5-15), and long (>15) clips

```
Results for 'Can you catch me up on the latest stories about Trump's tariffs?':

=== Short (0-5 min) ===
Title: Trump's Tariffs and Government Shutdown Updates (0.71)
Date: 2026-02-23 | Duration: 3m 20s
Tags: Trump tariffs, China, DHS shutdown, FEMA funding, mortgage rates, JPMorgan
Trump accounts, Potomac River sewage spill,
```

Audio Segment Search and Retrieval

3. Read transcript or playback audio for each segment

```
? Select a story to view the transcript or play audio: Short | Trump's Tariffs a
nd Government Shutdown Updates (3m 20s)
? Action for 'Trump's Tariffs and Government Shutdown Updates': (Use arrow keys)

 » ○ View Full Transcript
   ○ Play Audio Segment
   ○ Go Back
```

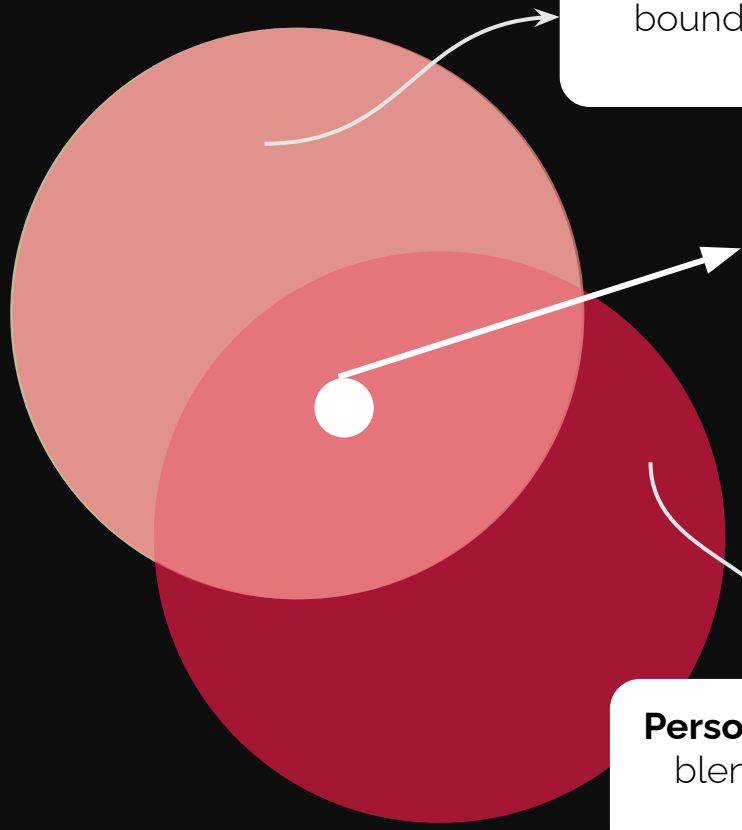
Accuracy: Man vs. Search

- Top 5 search results included **71%** of stories used in human playlist
- **95%** of retrieved stories were relevant to search query
- Next steps: use many searches and story segments to create a full playlist

Looking to the Future

What's Next?

Ethics: How do we draw boundaries for AI in the process?



The **ideal** sits in the center.

Solve Segmentation: Can a computer find where stories start and end?

Personalization: What's the right blend of personalization and randomization?

Thank You!

We would like to thank the following people for their contributions:
The ECJ teaching team (Steve, RB, Serdar, Maneesh, Lucy), Ethan and Bryan (KQED), Evan, Amine, Jason (B-Spoke), Sara (ex-NPR), Adam (Embarcadero) as well as all of our study participants.

Questions?